

# Eswar Adapa

## Data Scientist / AI Engineer

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[Website](#) | [GitHub](#) | [LinkedIn](#)

### PROFESSIONAL SUMMARY

Data Scientist and AI Engineer with 2+ years of experience in software engineering, system design, and end-to-end delivery of AI-powered systems. Builds scalable, production-grade solutions across Big Data Engineering, Machine Learning, Knowledge Graphs, and Agentic AI. Scaled a POI pipeline from 23M to 220M records, built a multi-agent internal operations platform, and integrated an 83M+ node Knowledge Graph with a LangGraph search agent. Received the Best Employee Award (February 2024 to May 2025).

### SKILLS

|                    |                                                                                                                                                                      |
|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Languages          | Python, C, SQL, Cypher                                                                                                                                               |
| ML & AI            | Machine Learning, Deep Learning, Natural Language Processing (NLP), Generative AI, Large Language Models (LLMs), Transformers (DistilBERT), Recommender Systems, OCR |
| Agentic AI         | LangGraph, LangChain, Retrieval-Augmented Generation (RAG), Graph RAG, Multi-Agent Systems, Tool/Function Calling, Prompt Engineering, Guardrails, Evaluations       |
| Big Data           | PySpark, Apache Sedona, Hadoop, Delta Lake, Apache Iceberg, Airflow, Trino, Parquet                                                                                  |
| Backend & APIs     | FastAPI, Flask, REST APIs, SSE Streaming, Pydantic, RabbitMQ, WebSockets, pandas                                                                                     |
| Databases          | Neo4j, MongoDB, Redis, MySQL, Qdrant, FAISS, ChromaDB                                                                                                                |
| Graph & Geospatial | Knowledge Graphs, Graph Data Science (GDS), Node2Vec, GraphSAGE, H3 Indexing, Spatial Joins                                                                          |
| Cloud & MLOps      | Amazon Web Services (AWS) (EC2, S3, EKS, SageMaker, Bedrock), Docker, Kubernetes, Jenkins, CI/CD                                                                     |
| Observability      | Langfuse, Datadog, New Relic, Loguru                                                                                                                                 |

### EXPERIENCE

Data Scientist / AI Engineer, Infinite Analytics February 2024 – Present

- POI Processing and Data Scaling
  - Consolidated POI crawling, ingestion, and validation pipelines using **RabbitMQ, Redis, MongoDB, and FastAPI**, expanding coverage from **23M POIs across 4 countries** to **220M POIs across 38 countries**.
  - Improved POI-type mapping accuracy from **83% to 86.7%** by replacing a 51-model **LightGBM** cascade with a unified **DistilBERT** model trained on **2.5M POIs**, achieving **0.005s** inference latency.
  - Reduced geospatial error rates from **18% to 14%** through plus-code mappings and triangular normalization.
  - Shortened Point-to-Polygon mapping runtime from **1 week to 2.5 hours** using **Apache Sedona** with a partition-aware pre-join strategy and **Apache Iceberg**.
- Sherlock Agentic Space: Multi-Agent Operations Platform
  - Migrated Sherlock AI from a monolithic **LangChain** Customer Success agent to a **LangGraph** multi-agent system with seven specialists, tool/function calling, memory and context management, and MongoDB persistent state, achieving **10x throughput, 31% lower token cost**, and reducing response latency from **8–12s to 3–5s**.
  - Expanded the platform into internal agents for Customer Success, Scrum, Sales, Marketing, HR, and Product operations through a unified **Flask** dashboard with persona-routed search and RAG retrieval.
  - Strengthened reliability with plan-aware guardrails, deterministic identifiers, and an overlay model that preserved manual edits across re-runs, contributing to **12% higher customer retention, 18% more qualified sales leads, and 15% growth** in the company LinkedIn channel.
- POI Knowledge Graph with Integrated LangGraph Search Agent
  - Constructed a **15-node-type, 14-relationship Knowledge Graph** over 83M+ POIs in **Neo4j**, enabling Graph RAG and reasoning across structured and unstructured enterprise data.
  - Generated graph embeddings using **Node2Vec, FastRP, HashGNN, and GraphSAGE** with Graph Data Science, achieving **8x speedup** in person resolution and **6–15x speedup** in POI ingestion workflows.
  - Integrated the graph with an autonomous **LangGraph** search agent supporting consumer discovery and B2B location intelligence through a hybrid Graph and Vector RAG architecture.
  - Delivered a **FastAPI** SSE streaming chat service with 22 tools, a clarify-first protocol for business decisions, and a sandboxed analytics engine; monitored systems using **Langfuse, Datadog, and New Relic**, and deployed through **Docker, Jenkins CI/CD, and Kubernetes**.

Research Intern, DRDO May 2023 – August 2023

- sEMG Signal Analysis and Gait Modeling
  - Trained Deep Learning models on IMU and sEMG data to classify seven gait phases, achieving **92% accuracy**.
  - Improved signal preprocessing pipelines, resulting in a **20% increase** in signal quality.
  - Applied Symbolic Transfer Entropy to analyze muscle interactions and identified a **25% increase** in inter-muscle information transfer associated with walking speed.

## PUBLICATIONS

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**Fatigue Classification and Onset Estimation Using sEMG Signals During Strength Training** August 2022 to April 2023  
*APSIPA'23 - IEEE*

- Presented at APSIPA'23 conference, Taiwan (IEEE); achieved **86% classification accuracy** using ML models with Majority Windowing and a **12% estimation error**.

## EDUCATION

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**Indian Institute of Information Technology, Sricity** December 2020 to April 2024  
*B.Tech.(Honors), Electronics and Communication Engineering*

- CGPA: 8.02

## ACHIEVEMENTS

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**Best Employee Award** February 2024 to May 2025

- Recognized for leading 2 of 3 high-impact projects that expanded business coverage from 4 to 38 countries, spanning Data Engineering, Knowledge Graph development, and GenAI integration.

**Academic Grand Challenge on Climate Change, Nasscom and Telangana Government** February 2023 to March 2023

- Ranked among top 40 teams out of 7,000+ participants; built a predictive framework for heatwave events and Air Quality Index in Tier-2 cities of Telangana.

## POSITION OF RESPONSIBILITY

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**Internship Manager - BITS Pilani Collaboration** June 2025 to July 2025

- Served as Scrum Master for 5 BITS Pilani interns, directing sprint planning, task allocation, and mentoring on technical practices.